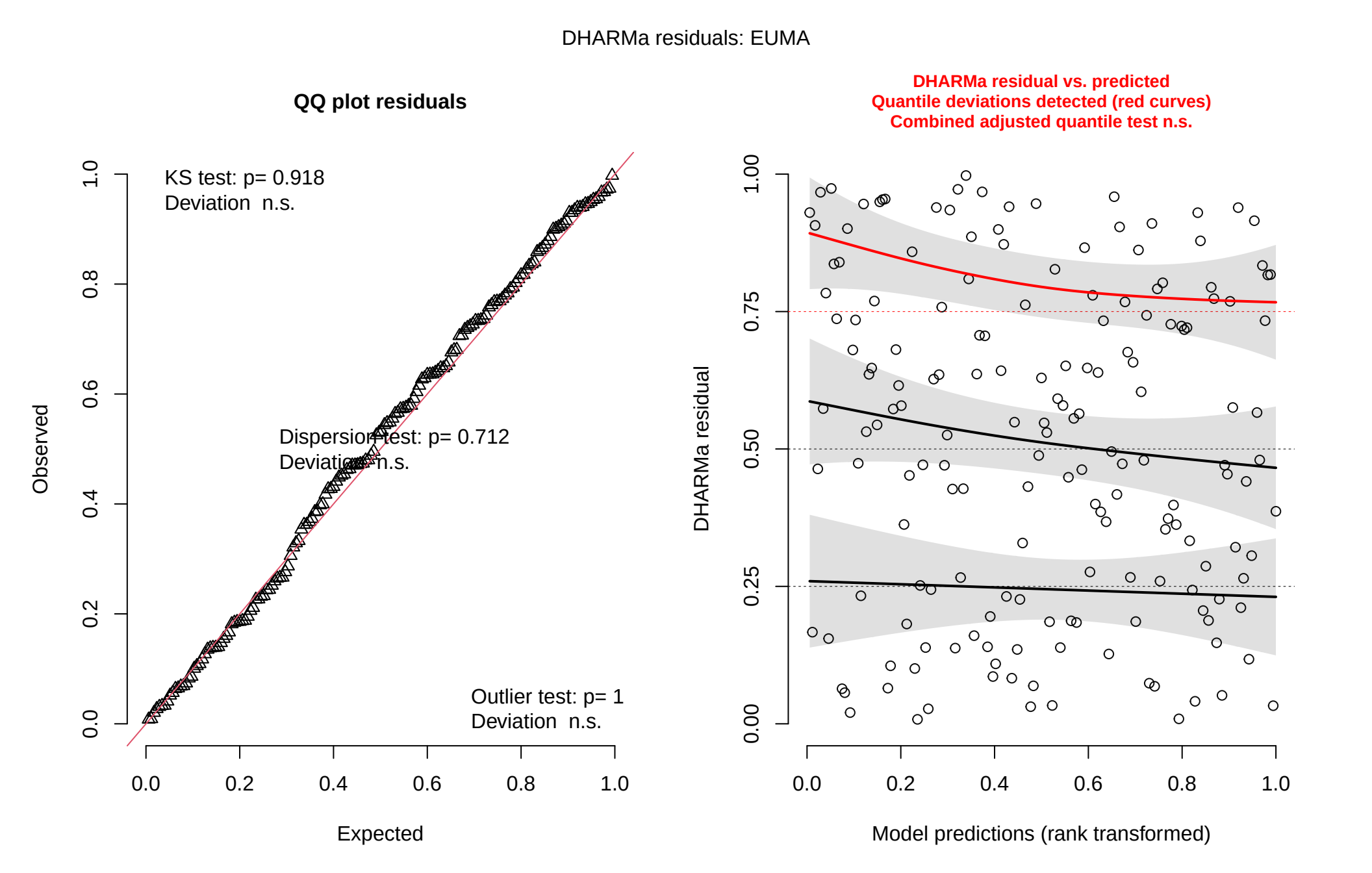
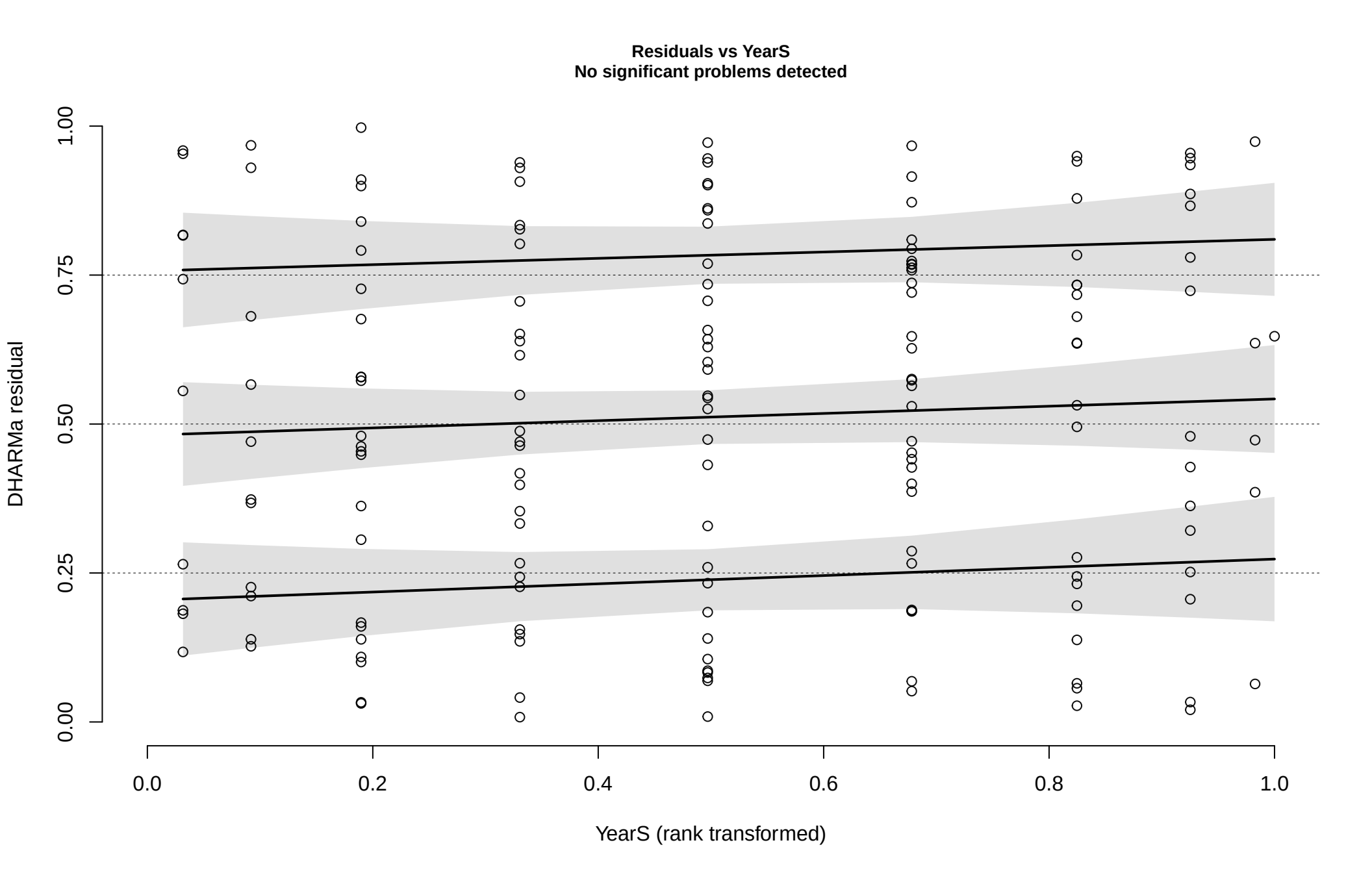
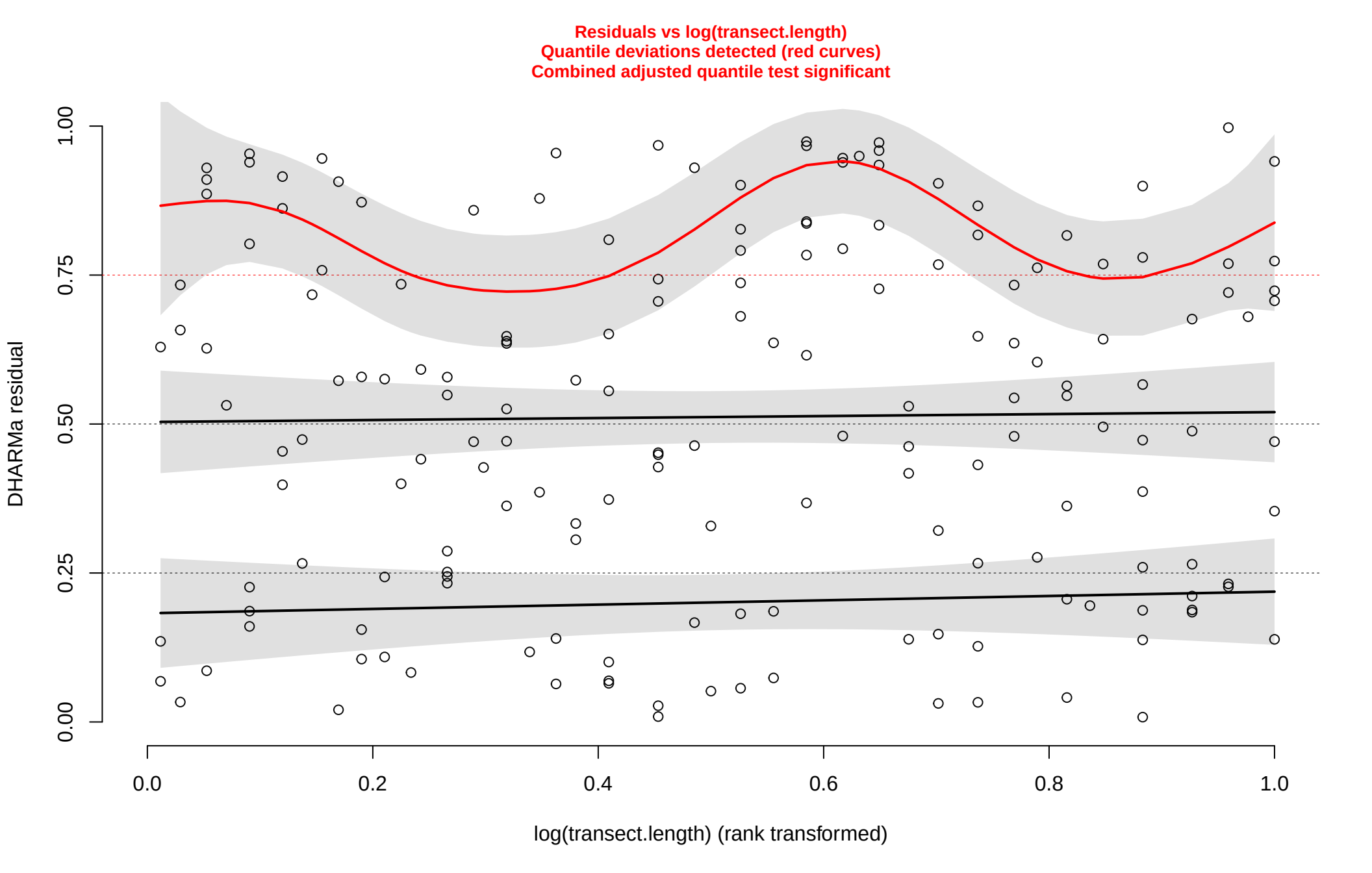
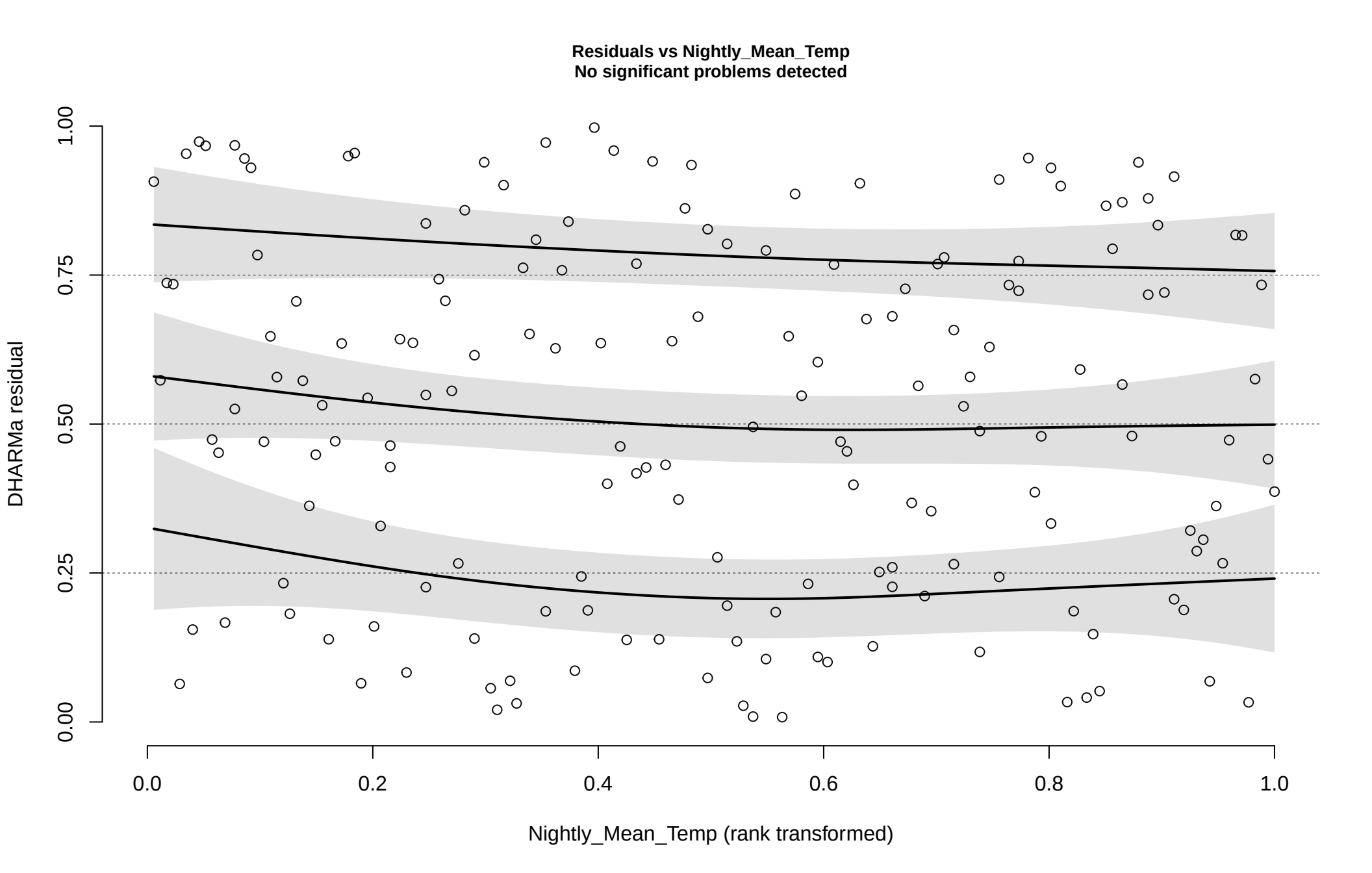
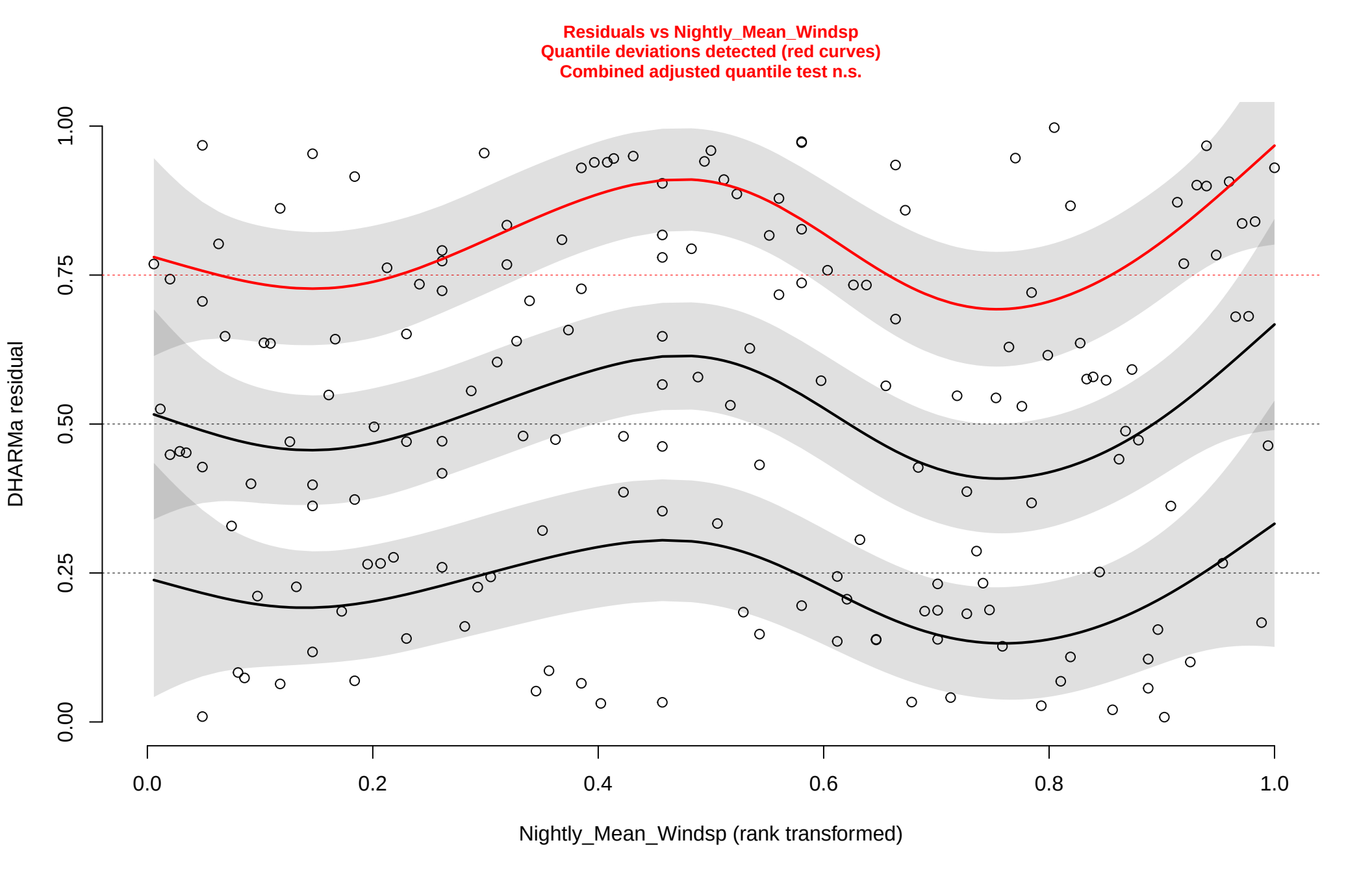
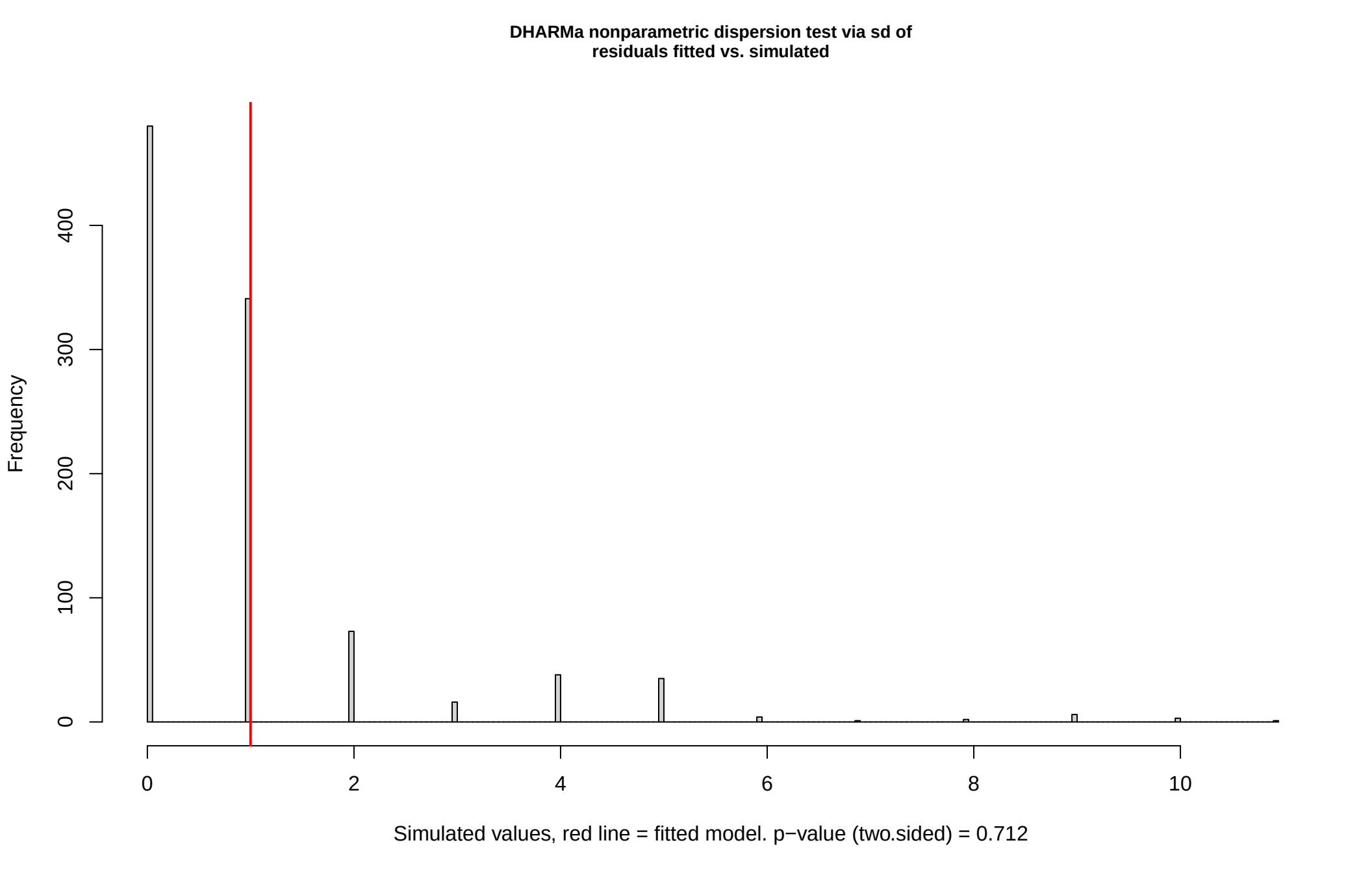

DHARMa residuals: EUMA
QQ plot residuals
KS test: p= 0.918
Deviation n.s.
Dispersion test: p= 0.712
Deviation n.s.
Outlier test: p= 1
Deviation n.s.
Observed
Expected
0.0
0.2
0.4
0.6
0.8
1.0
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
DHARMa residual
Model predictions (rank transformed)
0.00
0.25
0.50
0.75
1.00

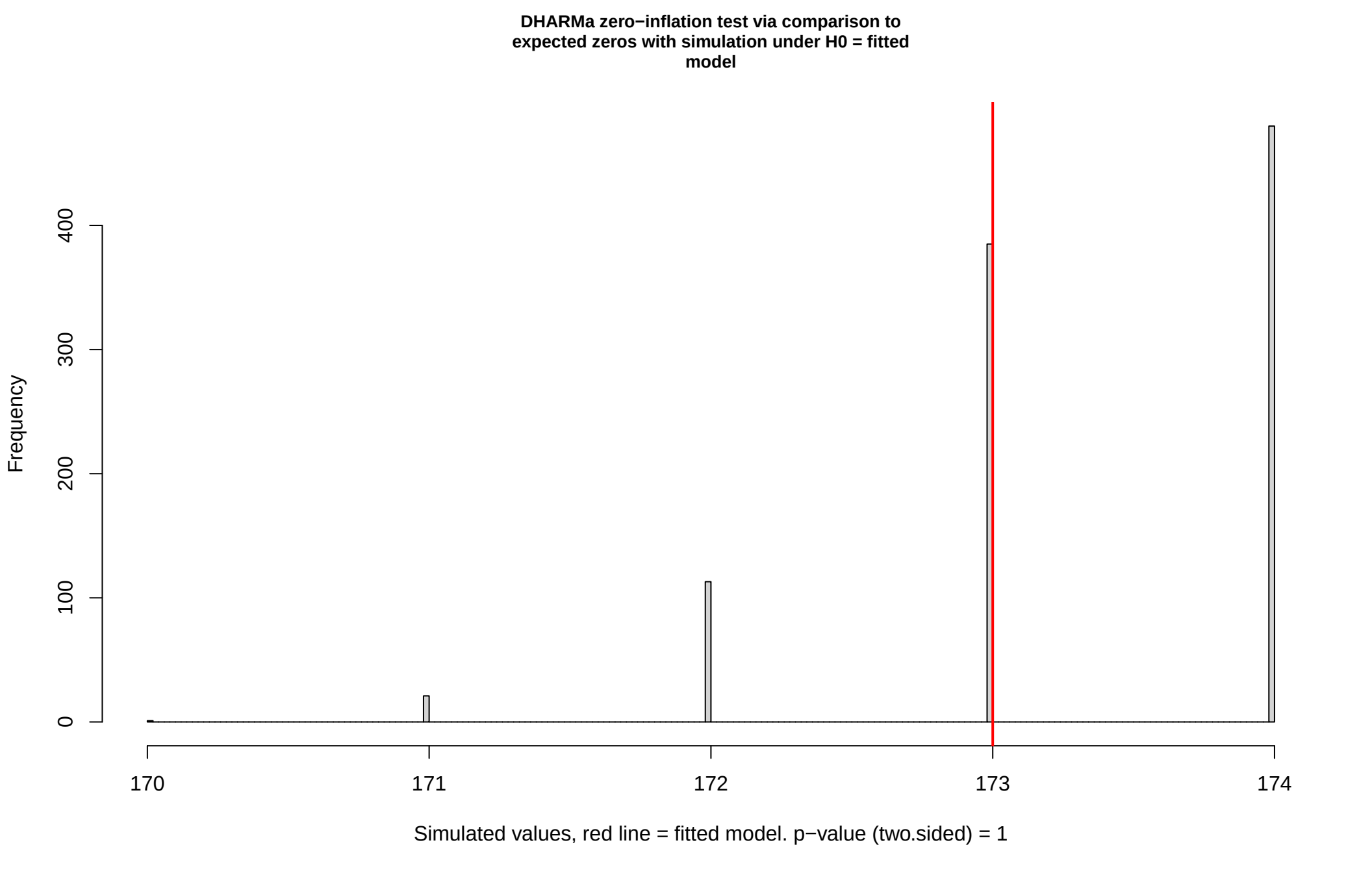

Residuals vs YearS
No significant problems detected
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
YearS (rank transformed)

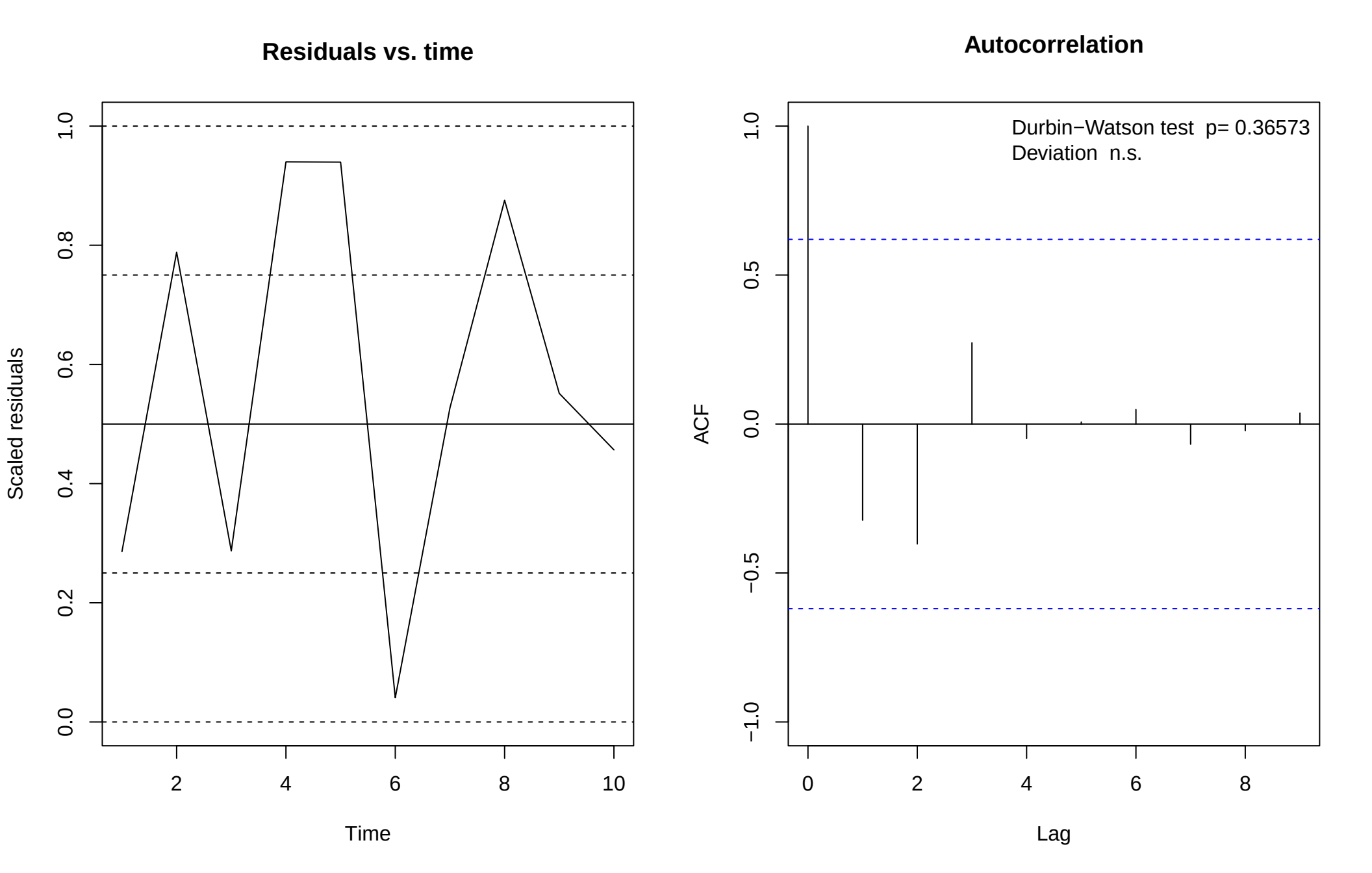

Residuals vs log(transect.length)
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
log(transect.length) (rank transformed)

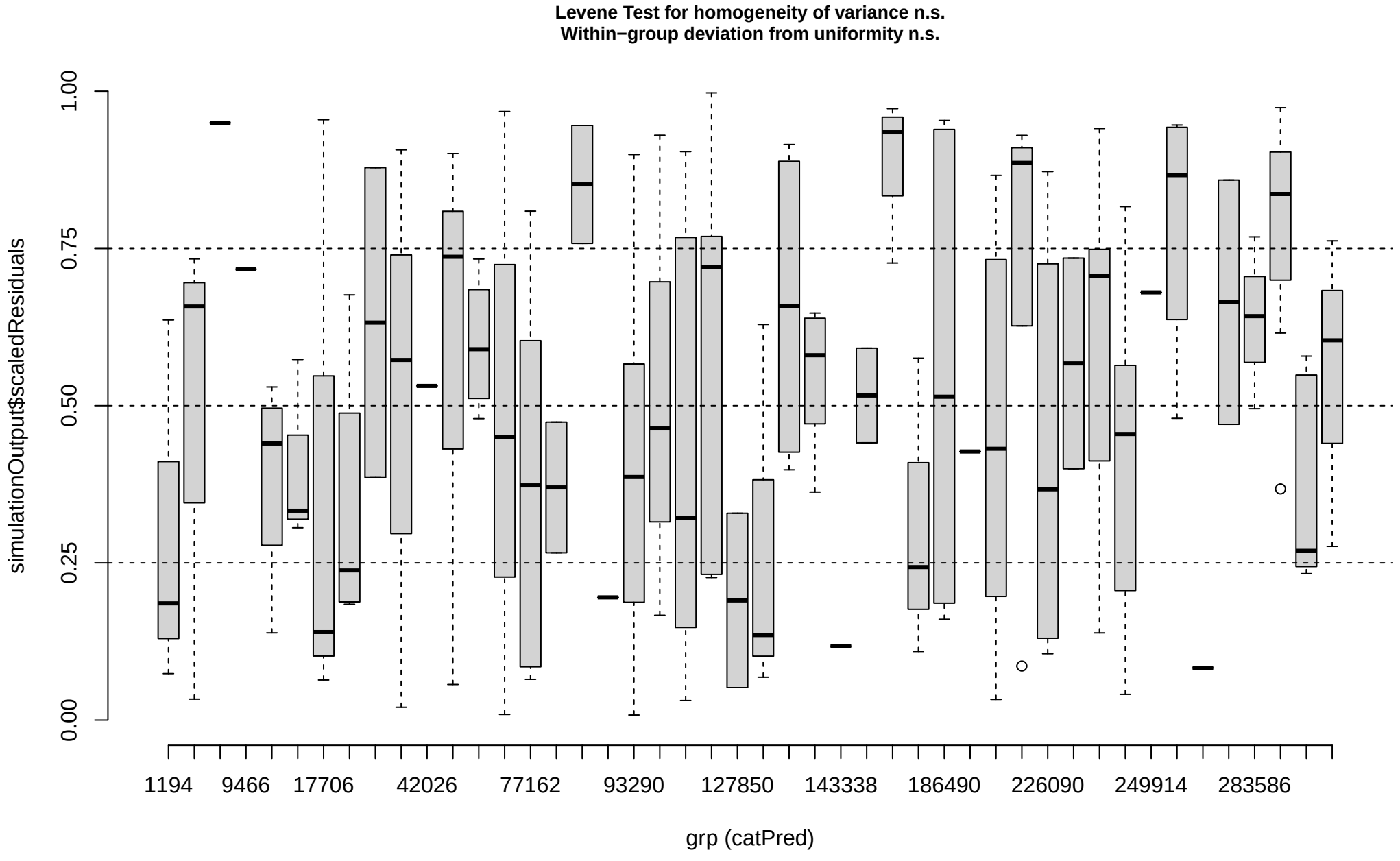

Residuals vs Nightly_Mean_Temp
No significant problems detected
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Temp (rank transformed)

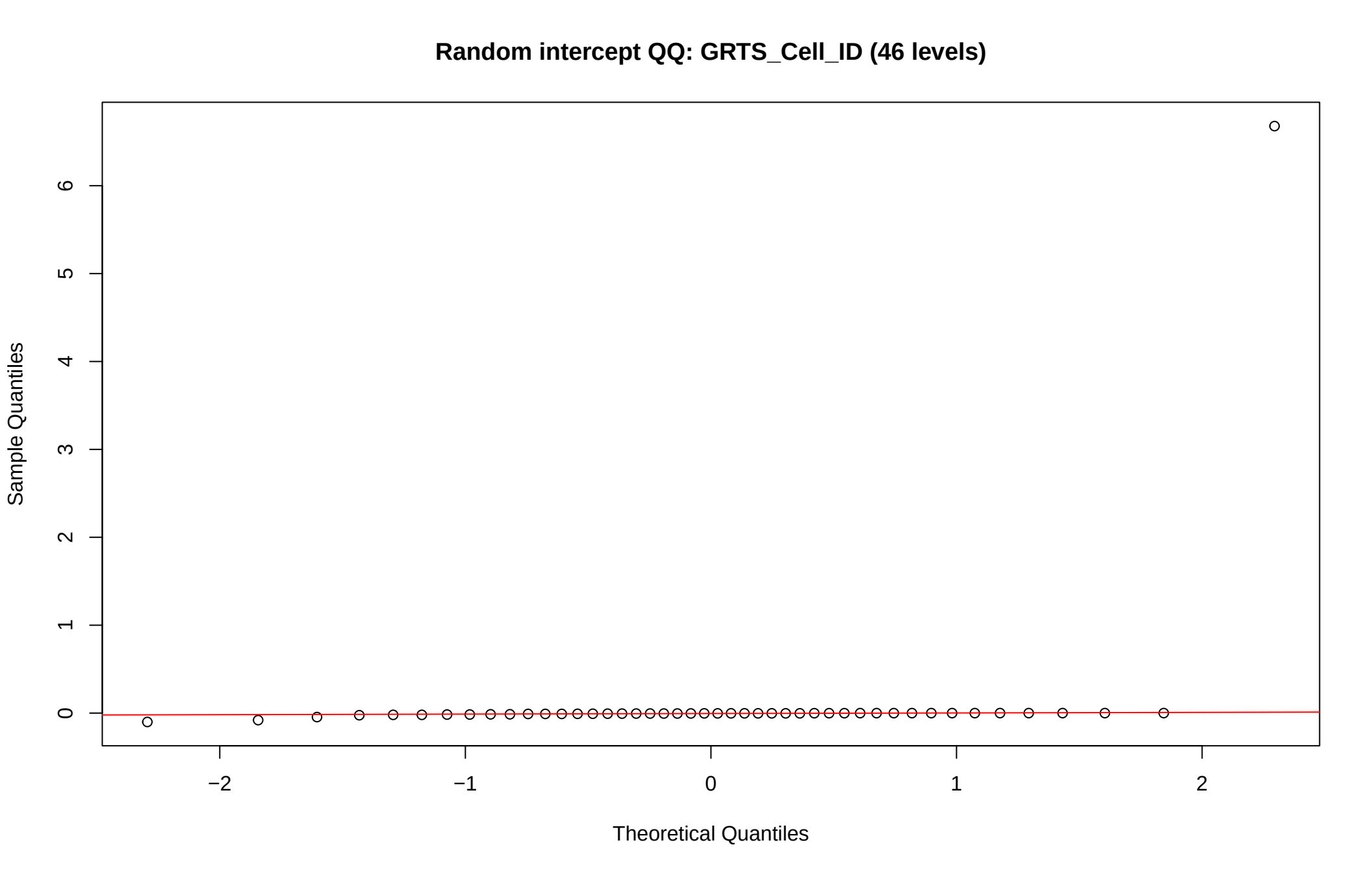

Residuals vs Nightly_Mean_Windsp
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Windsp (rank transformed)


DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
0
100
200
300
400
0
2
4
6
8
10
Simulated values, red line = fitted model. p−value (two.sided) = 0.712


DHARMa zero−inflation test via comparison to expected zeros with simulation under H0 = fitted model
Frequency
0
100
200
300
400
170
171
172
173
174
Simulated values, red line = fitted model. p−value (two.sided) = 1


Residuals vs. time
Scaled residuals
0.0
0.2
0.4
0.6
0.8
1.0
2
4
6
8
10
Time
Autocorrelation
ACF
−1.0
−0.5
0.0
0.5
1.0
Durbin−Watson test p= 0.36573
Deviation n.s.
0
2
4
6
8
Lag


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
0.00
0.25
0.50
0.75
1.00
1194
9466
17706
42026
77162
93290
127850
143338
186490
226090
249914
283586
grp (catPred)


Random intercept QQ: GRTS_Cell_ID (46 levels)
Sample Quantiles
Theoretical Quantiles
0
1
2
3
4
5
6
−2
−1
0
1
2